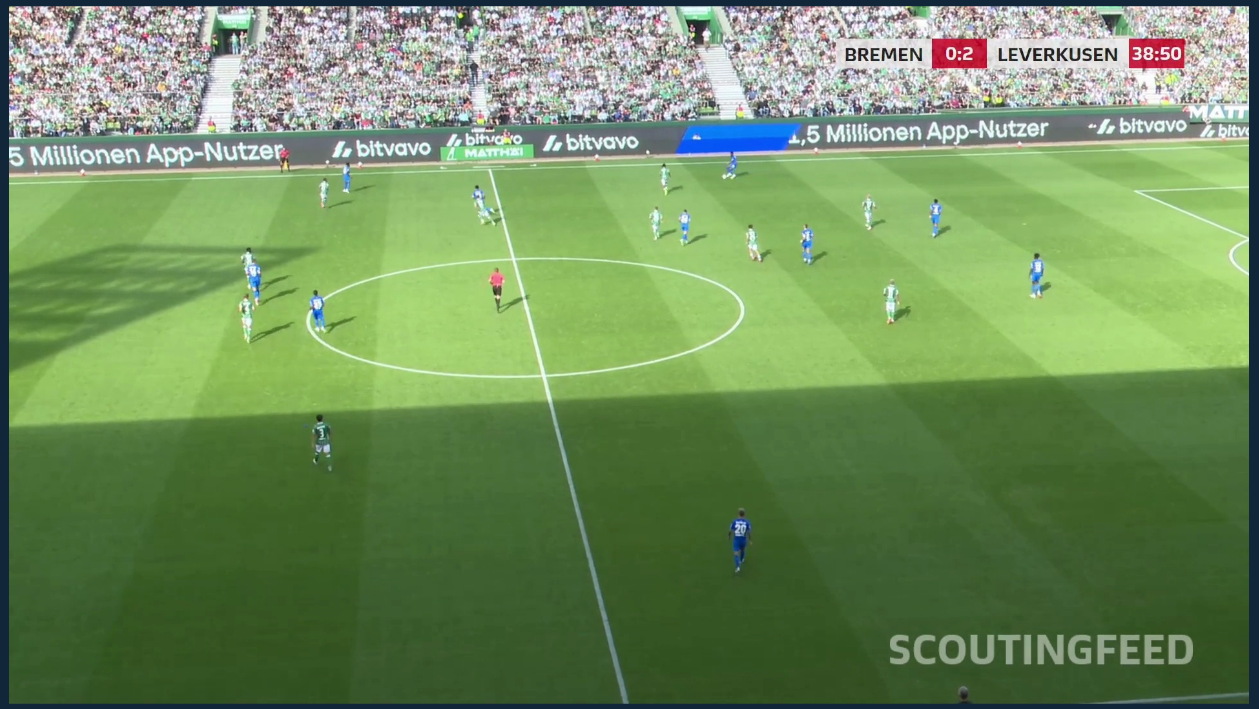
Pose estimation in football video

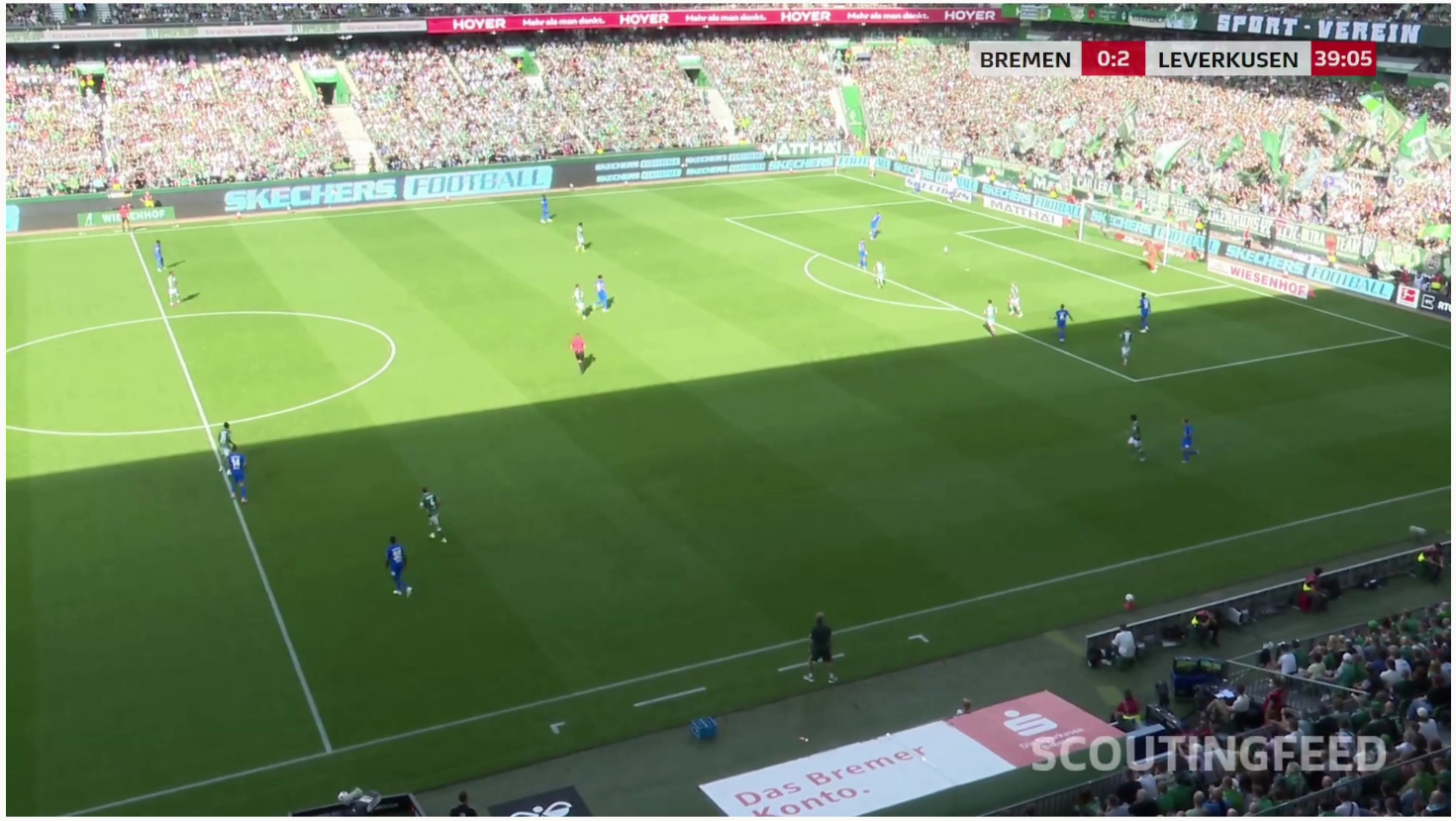
Preprocessing experiments
and current project status



Master’s thesis progress review
23 October 2026

Research question

Which preprocessing steps help pose estimation in wide-angle football video?



The challenge

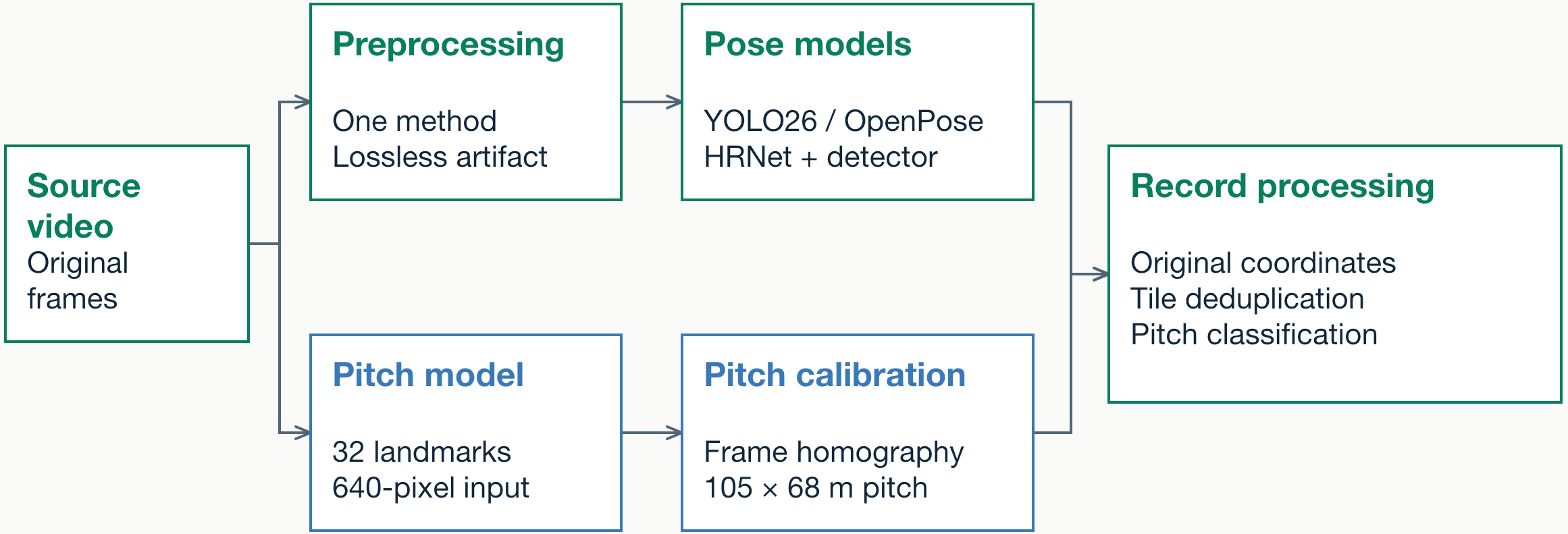
Small players
Strong shadows
Crowd and sideline detections
Moving camera

Research objective

Find a reproducible pipeline for reliable player poses.

Pipeline structure

Pose inference and pitch calibration use separate branches of the same source video.



Outputs: raw and on-pitch poses, decision tables, metrics and annotated videos

Model setup

Standard pose checkpoints stay fixed across preprocessing conditions.

Model	Pose approach	Football inference setup
YOLO26x Pose	Multi-person pose	1920 target size
OpenPose BODY_25	Bottom-up	1920 × 1088 network input
HRNet-W32	Top-down	Football YOLO detector + 256 × 192 person crops

Common output: COCO-17 body-keypoint layout in original-frame coordinates

The player detector and the YOLOv8x pitch model are separate auxiliary models.

Single-step experimental design

One source clip and fixed model settings isolate each processing change.

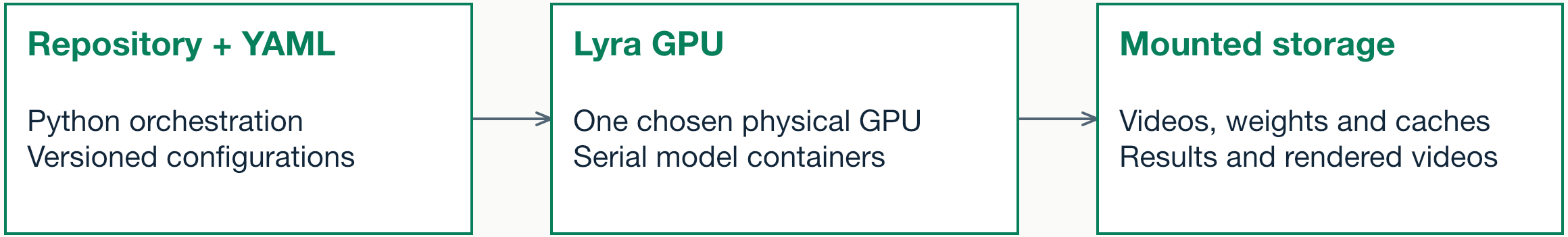
12 conditions × 3 models = 36 jobs

Family	Conditions
Reference	Unchanged baseline
Scale and regions	Bicubic ×2, super-resolution ×2, tiling, detector-guided crop
Contrast and brightness	CLAHE, gamma darken, gamma brighten
Noise and blur	NLM denoise, bilateral denoise, unsharp, motion deblur

Sample: 782 frames, 1920 × 1080, approximately 30 seconds

Lyra execution and reproducibility

The code, model runtimes and large experiment files have separate roles.



Run control

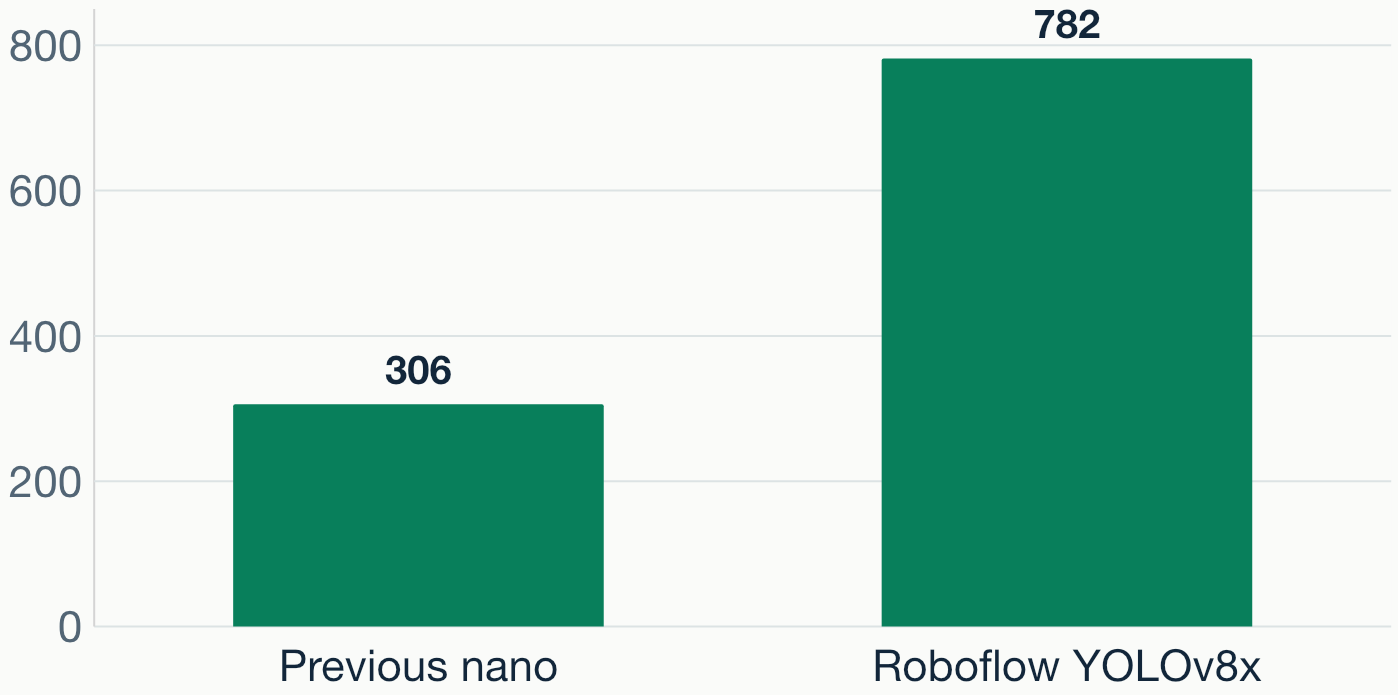
tmux keeps the run alive
--device selects the GPU
Preflight checks dependencies

Audit trail

Checkpoint and input hashes
Expanded YAML, per-job logs and JSON summaries
Raw and filtered Parquet, comparison tables, MP4s

Roboflow improved pitch-fit coverage

Controlled pitch-only comparison on the same 782-frame video



39.1% to 100%

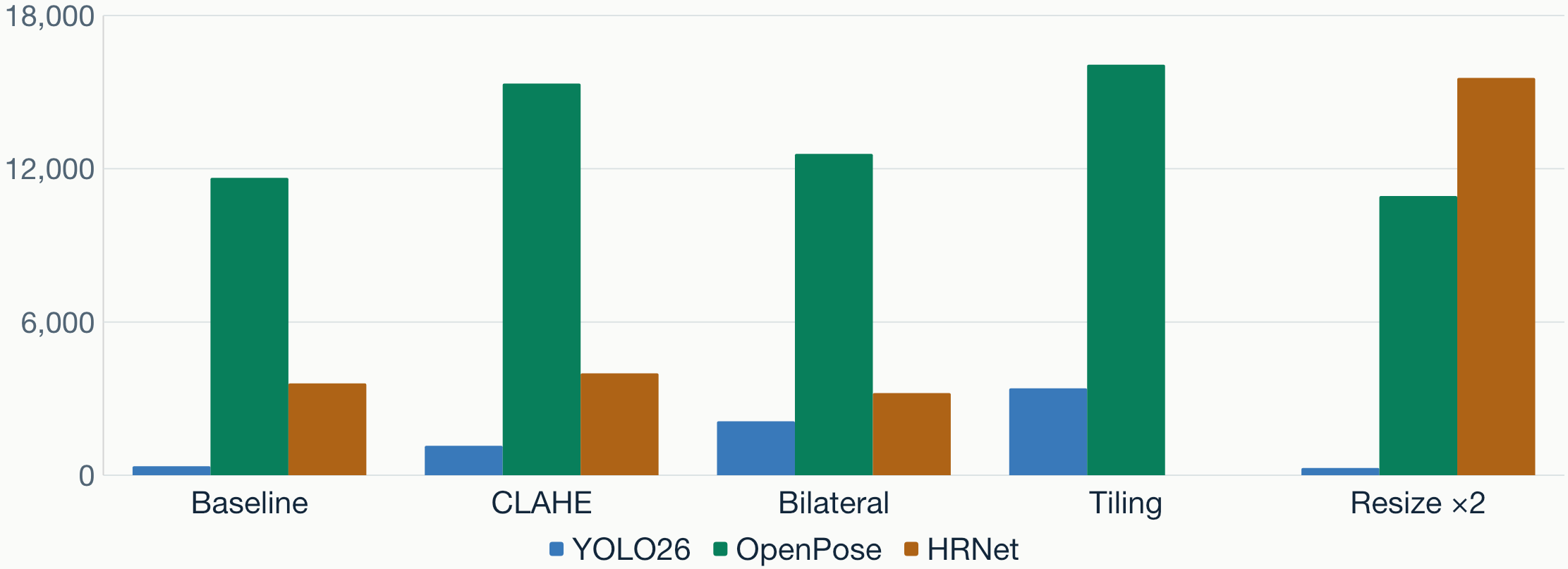
Accepted geometry coverage

Shared 640 input
Shared fit checks
105 × 68 m Bremen pitch
No frame-to-frame fallback

Fit acceptance measures geometric consistency. It does not establish ground-truth landmark accuracy.

Processing effects depend on the model

On-pitch records over 782 frames, latest screening from 22 October
Detection yield only. Ground-truth accuracy evaluation is pending.

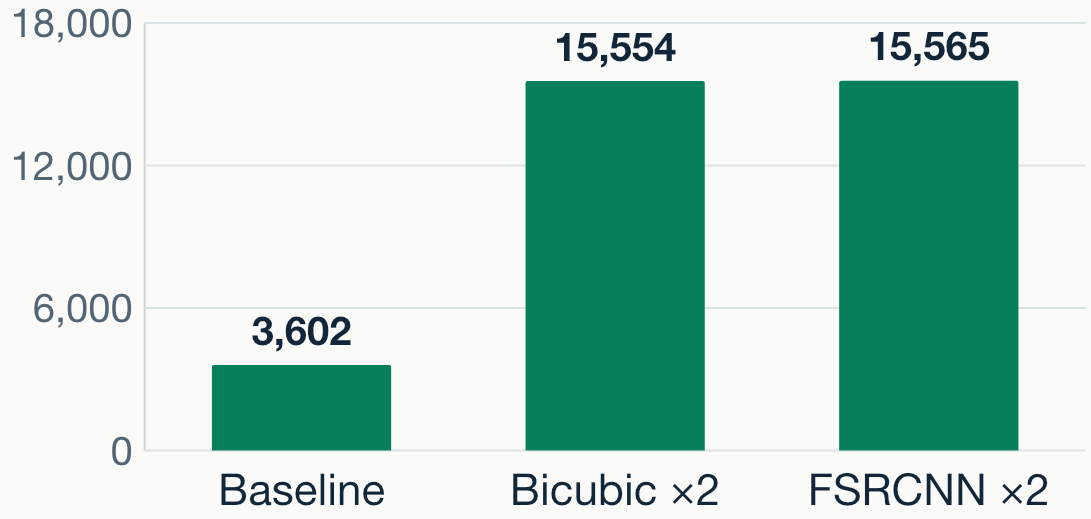


OpenPose + CLAHE: +31.7% **YOLO26 + bilateral: +492.4%** **HRNet + resize: +331.8%**

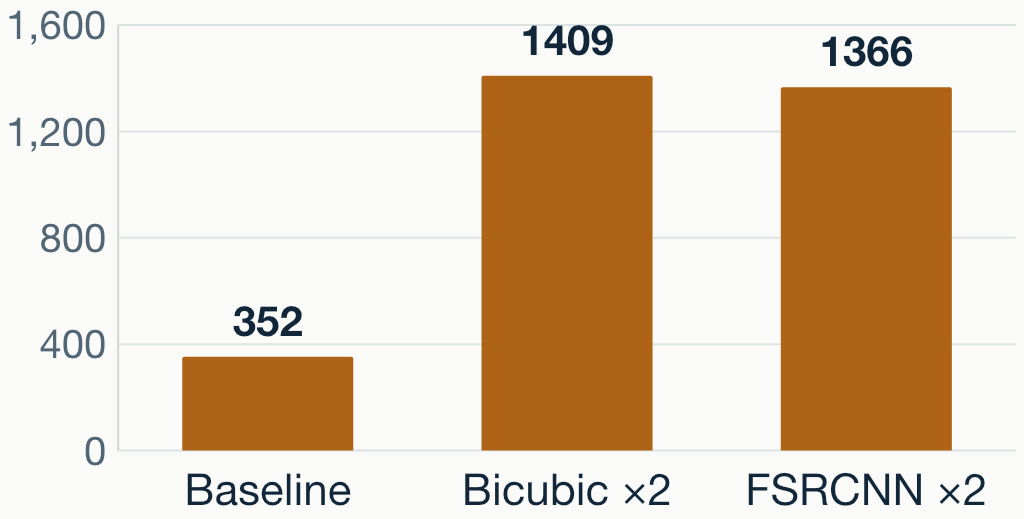
HRNet: resizing explains almost all of the SR yield

The matched bicubic control is essential for interpreting super-resolution.

On-pitch records



Model wall time, seconds



SR adds 11 records over bicubic, approximately 0.07%.

Both cost roughly 4× baseline model time. Cached preprocessing hides the actual processing cost.

Current status and unresolved issues

35 of 36 model jobs completed. Three model/processing cases need investigation.

Case	Observed result	Interpretation
HRNet + tiling	1 raw pose record	Suspicious integration behaviour
YOLO26 + cropping	0 raw pose records	Completed job, unusable yield
OpenPose + cropping	FAILED	Detailed error log still needed

Raw counts already show these problems before pitch filtering.

Root causes remain unconfirmed. These cases should not determine the final preprocessing ranking.

Next experiment stage

1 **Resolve the integration cases**

Targeted crop and tile reruns, preserving successful results

2 **Establish pose accuracy**

Annotated football frames and a documented COCO setup check

3 **Compare promising combinations**

Keep controls for scale, confidence and total runtime

4 **Connect poses to tracking data**

Pitch-coordinate player positions and a later 2D radar view

Discussion: annotated frame sample and primary accuracy metric

Backup: complete on-pitch overview 1/2

Each cell shows the record count and change relative to that model's unchanged baseline.

Processing	YOLO26	OpenPose	HRNet
Baseline	357 (reference)	11,647 (reference)	3,602 (reference)
Resize ×2	283 (-20.7%)	10,934 (-6.1%)	15,554 (+331.8%)
CLAHE	1,156 (+223.8%)	15,338 (+31.7%)	3,995 (+10.9%)
Gamma darken	315 (-11.8%)	12,193 (+4.7%)	3,634 (+0.9%)
Gamma brighten	440 (+23.2%)	11,029 (-5.3%)	3,568 (-0.9%)
NLM denoising	615 (+72.3%)	8,054 (-30.8%)	3,037 (-15.7%)

782 source frames. Yield diagnostics only. HRNet tiling and YOLO cropping need investigation.

Backup: complete on-pitch overview 2/2

Each cell shows the record count and change relative to that model's unchanged baseline.

Processing	YOLO26	OpenPose	HRNet
Bilateral denoising	2,115 (+492.4%)	12,581 (+8.0%)	3,216 (-10.7%)
Unsharp	491 (+37.5%)	11,860 (+1.8%)	2,675 (-25.7%)
Motion deblur	1,627 (+355.7%)	7,362 (-36.8%)	214 (-94.1%)
Super-resolution ×2	312 (-12.6%)	11,583 (-0.5%)	15,565 (+332.1%)
Tiling	3,402 (+852.9%)	16,072 (+38.0%)	1 (-100.0%)
Detector crop	0 (-100.0%)	FAILED	3,467 (-3.7%)

782 source frames. Yield diagnostics only. HRNet tiling and YOLO cropping need investigation.

Backup: COCO setup validation

Completed historical run on all 5,000 COCO val2017 images

Historical model	Keypoint AP	AP50	AR
YOLOv8x Pose	0.684	0.893	0.755
OpenPose BODY_25	0.296	0.650	0.343

These YOLO results belong to YOLOv8x, not YOLO26.

The YOLO26 benchmark configuration exists. Its result is not included in the supplied evidence. HRNet needs a separately defined COCO bounding-box protocol.